Autoballast

Autoballast is the Slocum software feature that automatically tunes how much buoyancy drive the glider uses, so a pilot only has to specify the *total* drive they want and let the glider find the rest. It does two things:

1. **Reduces the drive** used on dives and climbs to the smallest amount that still flies the glider acceptably — saving energy and reducing speed.
2. **Centers** that drive so the dive and climb are balanced (a dive/climb speed ratio near **1:1**) and the pump does **not** have to fully extend at the surface every cycle.

It is especially valuable on **G3** gliders, whose high-displacement pumps move a lot more volume — running them at full `±1000 cc` drive "significantly increases energy consumed" and can cause fast, steep, uncontrolled dives in shallow water. See Power Saving, which recommends autoballast as a core energy-conservation lever.

> **Source**
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> Paraphrased from the `masterdata` autoballast/speed-control sensor block and the *Slocum G3 Glider Operators Manual* (the public `doco/how-it-works/autoballast.txt`), with field practice from the Teledyne Webb Research user forum and the UG2 community Slack. The concept — the glider "adjusts the center point of the drive to provide a dive/climb ratio of one" given a user-defined total volume — is described in TWR's UUST 2013 Slocum paper. Defaults and sensor names vary by firmware; **confirm against your `masterdata` and simulate before flying.**

What it does

Without autoballast you command a fixed buoyancy: full drive is `±1000 cc`, but a glider flies fine on far less (often **~300 cc total**, sometimes less). Picking that reduced drive by hand — and re-centering it as ballast, water density, and payload change — is fiddly. Autoballast does it continuously:

- You give it **one number**: the **total** drive volume you want (the spread between the dive and climb buoyancy). There is **no separate "climb" value** — the single total is split around a center point the glider chooses.